The Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory generates 20 terabytes of astronomical imagery nightly, enabling precision cosmological measurements of dark energy and dark matter. Image quality depends on mitigating dome seeing, a form of wavefront distortion caused by lateral temperature gradients in the telescope enclosure. Correcting this thermal structure to 0.1°C would bring the point spread function within 10% of the diffraction limit.

The Bidirectional Acoustic Traveltime Pulser for Index Gradients (BATPING) measures path-integrated air temperature via round-trip travel of ultrasonic pulses without disrupting normal telescope operation. Bidirectional propagation cancels airflow contributions to the measured speed of sound, isolating the thermal signal. Piezoelectric transducers at 40 kHz on a 10-meter baseline would transmit and receive waveforms tailored to the transducers' nonlinear resonant response. Laboratory validation employs a 2-meter benchtop acoustic path with transducers driven by an arbitrary waveform generator and digitized at 384 kHz.

After evaluating different waveform parameterizations using a Bayesian optimization algorithm, we found that an amplitude-compensated linear frequency-modulated chirp, with an envelope shaped by the inverse of the measured piezoelectric transfer function, maximizes signal quality. A signal-processing pipeline implementing adaptive window slicing for matched filtering and a full-waveform least-squares shape fit was also developed toward the aim of precision measurement.

Our preliminary results demonstrate a per-ping temperature precision of 0.045°C on a 2-meter path, surpassing the 0.1°C target by more than a factor of two, with a traveltime estimation precision of approximately two nanoseconds per measurement.

Future work includes the design of a transponder circuit that actively retransmits the pulse after a known delay to eliminate signal attenuation at dome-scale path lengths. We anticipate deployment at the Rubin Auxiliary Telescope dome before the end of 2026, with the longer-term objective of two-dimensional acoustic tomography of the full dome temperature structure.